

Intelligent Ambulance Dispatch and Real-Time Coordination for Emergency Response

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Abstract - Getting help quickly during a medical emergency can significantly affect a patient's chances of survival, especially within the first hour after the incident. In growing cities, emergency response is often delayed by traffic, uneven ambulance availability, and poor communication among the patient, ambulance provider, and hospital. This survey reviews existing research on ambulance dispatch, hospital selection, driver-side coordination, communication infrastructure, and live tracking for emergency response. Most existing systems either select ambulances primarily based on proximity without fully considering real-time traffic, or address only one part of the process, such as routing or hospital selection, rather than the complete response chain. Few systems automatically identify an alternative ambulance when the initially assigned ambulance cannot respond, or maintain continuous real-time coordination among the patient, driver, and hospital. These limitations highlight the need for a unified, real-time coordination platform that integrates ambulance selection, fallback dispatch, and live tracking within a single system. To address this gap, UyirKappan presents a unified emergency-response platform with a working prototype demonstrating intelligent ambulance matching, cascading fallback dispatch, and real-time multi-stakeholder tracking.

Index Terms - Ambulance Dispatch, Emergency Medical Services, Real-Time Tracking, Healthcare Coordination, Urban Emergency Response.

I. INTRODUCTION

Timely emergency medical response plays a critical role in improving patient outcomes during trauma, cardiac emergencies, and other acute conditions. The Golden Hour, the early period following a critical medical event, is widely recognised as a period during which timely medical intervention can have a significant impact on survival. Despite this, urban emergency response systems continue to face challenges such as traffic congestion, uneven ambulance distribution, and delays in communication among bystanders, ambulance providers, and hospitals. In many cases, the nearest ambulance is not necessarily the fastest to reach the patient, as road conditions and route differences can significantly affect travel time.

Existing research has addressed several aspects of this problem, including ambulance dispatch, hospital selection, driver-side coordination, backend infrastructure, live tracking, and estimated time of arrival. However, these solutions often address individual components rather than integrating them into a single coordinated system. Dispatch methods frequently rely on geographic proximity without fully accounting for traffic and route conditions. Hospital-selection studies primarily focus on destination choice rather than continuous coordination across the complete emergency response chain. As a result, coordination among the bystander, driver, and hospital may remain fragmented, while recovery from failed ambulance assignments is often not addressed.

This survey examines existing research on ambulance dispatch, real-time coordination, and cascading fallback for emergency response. The reviewed works are organised by functional area, and a comparative summary is provided to identify their contributions and limitations. The survey highlights a gap in the integration of traffic-aware ambulance selection, automatic reassignment when an assigned ambulance cannot respond, and continuous real-time coordination across emergency stakeholders. These findings motivate the need for a unified emergency-response platform such as UyirKappan.

II. RELATED WORK

The growing demand for faster emergency medical response has led to research in ambulance dispatch, hospital coordination, backend infrastructure, intelligent matching, dynamic redeployment, and ETA estimation. These studies address different components of emergency response, but they vary considerably in scope, level of integration, and real-time coordination capability.

Rafaqat et al., 2025 [1] present EMCON, a digital emergency-response platform connecting patients, ambulances, and hospitals. The system demonstrates the value of real-time resource sharing and stakeholder coordination, but does not provide integrated intelligent ambulance matching and traffic-aware dispatch.

Smith et al., 2022 [2] evaluate the GoodSAM volunteer first-responder application for connecting nearby trained responders with cardiac-arrest incidents. The study demonstrates the potential of mobile and location-based emergency assistance, although its scope is limited to volunteer first response and cardiac arrest.

Yang et al., 2024 [3] investigate personalized dispatch mechanisms that account for driver acceptance behaviour in on-demand mobility services. While relevant to assignment and driver-response decisions, the approach is designed for ride-hailing and does not address the specific requirements of emergency ambulance dispatch.

Fu et al., 2022 [4] develop a simulation framework for studying ambulance dispatch and hospital decision-making under capacity and travel-time constraints. The work demonstrates the interaction between ambulance operations and hospital decisions, but remains primarily a modelling and simulation approach rather than an operational real-time platform.

Olivier et al., 2022 [5] develop a probabilistic approach for recommending hospitals to EMS vehicles using ambulance and vehicle-telematics data. The study demonstrates the importance of travel-time uncertainty in hospital selection, but focuses mainly on destination recommendation rather than end-to-end emergency coordination.

Xu et al., 2024 [6] propose a simulation-based hospital recommendation system that considers future hospital bed availability together with routing information. The work highlights the importance of incorporating hospital capacity into emergency destination selection, although its evaluation is based on simulation and historical data.

Zaki et al., 2022 [7] investigate a cloud-assisted microservice architecture for healthcare applications. Their work demonstrates how distributed service architectures can support scalable healthcare systems and provides architectural principles relevant to the backend coordination requirements of emergency-response platforms.

Ankarboina et al., 2026 [8] propose RACER, a congestion-aware emergency-routing framework that dynamically adapts routes according to changing road conditions. The study is relevant to traffic-aware emergency routing, but primarily addresses the routing of an individual emergency vehicle rather than ambulance resource allocation.

Xu and Li, 2025 [9] propose a dynamic risk-field approach for predicting urban traffic congestion using multiple sources of traffic and contextual information. The work demonstrates how spatial congestion information can support proactive traffic-aware decision-making in emergency transportation.

Nozari et al., 2025 [10] combine machine learning with Genetic Algorithm-based optimization for ambulance routing under uncertain conditions. The study demonstrates the potential of intelligent optimization for reducing ambulance

response and transportation costs, while addressing routing primarily as an optimization problem.

Abdeen et al., 2022 [11] propose a smart ambulance decision-making framework that jointly considers ambulance response, road congestion, and hospital loading. The work closely relates to intelligent emergency dispatch because it considers both the journey to the patient and the subsequent hospital destination, but is primarily presented as a mathematical and simulation-based decision model.

Schmid, 2012 [12] investigated dynamic ambulance dispatch and relocation using Approximate Dynamic Programming under stochastic conditions. The study demonstrates the importance of anticipating future demand and ambulance availability when making dispatch decisions.

Tulsi et al., 2025 [13] propose a Deep Reinforcement Learning approach for proactive redeployment of critical-care response vehicles. The work demonstrates how learned policies can adapt ambulance positioning to changing demand, but its primary focus is redeployment rather than failure recovery for an active emergency request.

Chu et al., 2023 [14] propose a deep-learning-based approach for improving estimated time of arrival prediction. The study highlights the potential of learning complex relationships among transportation-related features to improve ETA estimation.

Abid et al., 2024 [15] develop a machine-learning-based framework for estimating ambulance travel time from spatiotemporal data. The study demonstrates the value of machine-learning models for ambulance travel-time prediction, but primarily focuses on estimation rather than integrating ETA into a complete emergency-response workflow.

III. COMPARATIVE ANALYSIS

No	Author & Year	Inference	Disadvantages
1	Rafaqat et al., 2025 [11]	This work presents EMCON, a digital emergency-response platform that connects patients, ambulances, and hospitals. It offers real-time resource availability, ambulance coordination, inter-hospital transfers, and stakeholder communication for low- and middle-income settings	EMCON lacks an intelligent ambulance matching algorithm, allows patients to select ambulances based on preference, and does not provide ML-based ambulance selection or traffic-aware route optimization.

2	Smith et al., 2022 [2]	This study evaluates the GoodSAM volunteer first-responder app, integrated with ambulance CAD systems. It automatically alerts nearby CPR-trained volunteers during OHCA and enables pre-EMS intervention using smartphones and GPS.	Alert acceptance rates are approximately 1–5%, many OHCA incidents lack nearby responders, and the system focuses on cardiac arrest rather than general ambulance dispatch.
3	Yang et al., 2024 [3]	This work proposes personalized dispatch-menu optimization for on-demand mobility using probabilistic modelling of driver acceptance and decline behaviour. The approach reduces unmatched orders and improves overall driver utilization.	The approach is designed for ride-hailing rather than emergency medical services and does not consider patient severity, ambulance capabilities, hospital suitability, or emergency response time.
4	Fu et al., 2022 [4]	This paper proposes a coupled Game-Theoretical and Discrete-Event Simulation framework that models ambulance dispatch and hospital decisions using hospital capacity, travel time, and queuing.	The model uses simplifying assumptions, represents hospital behaviour mainly through Accept/Redirect strategies, and does not implement a real-time operational dispatch platform.
5	Olivier et al., 2022 [5]	This study develops a probabilistic hospital recommendation system using FDNY EMS historical data and vehicle telematics. It accounts for uncertainty in travel-time estimation when recommending hospitals.	The hospital suggestion pattern is largely static and does not fully capture temporal variation. The work primarily focuses on hospital selection rather than complete ambulance dispatch and coordination.

6	Xu et al., 2024 [6]	This paper proposes a simulation-based EMS hospital recommendation system using DES and Google Maps. It predicts future ED and ICU bed availability for 20, 40, and 60 minutes.	The system was not tested in real-time operational conditions, prediction accuracy varies between hospitals, and validation relies on historical and simulated data.
7	Zaki et al., 2022 [7]	This work introduces a Cloud-Assisted Microservice-Based Healthcare Application framework using DevOps and cloud deployment. It improves scalability, loose coupling, interoperability, and independent deployment.	Microservices introduce architectural complexity, defining appropriate service boundaries can be difficult, and no specific AI or optimization algorithm is proposed.
8	Ankarboina et al., 2026 [8]	This paper proposes RACER, a real-time congestion-aware emergency-routing framework using multi-edge look-ahead and congestion-aware Dijkstra. It avoids unnecessary rerouting while limiting communication overhead.	The framework requires real-time edge-level traffic information via V2I or telemetry and focuses on routing a single emergency vehicle rather than multi-ambulance resource allocation and assignment.
9	Xu & Li, 2025 [9]	This work proposes a Dynamic Risk Field Model for real-time urban traffic-congestion prediction. It combines traffic data, historical patterns, road topology, incidents, and weather into a spatial congestion-risk field.	Performance depends on the quality and coverage of traffic sensor or probe data, Gaussian-kernel parameters require calibration, and complex incidents are simplified through Gaussian decay.

10	Nozari et al., 2025 [10]	This paper designs an ambulance-routing optimization model using neural-network-based learning combined with Genetic Algorithm. It aims to reduce ambulance response time and transportation costs while enabling more emergency cases to be handled.	The model considers a limited set of variables and does not incorporate important real-time factors such as traffic congestion, ambulance availability, hospital capacity, and road conditions in the stated dataset.
11	Abdeen et al., 2022 [11]	This paper proposes a smart ambulance system jointly considering ambulance response, road congestion, and hospital loading across both the ambulance-to-patient and patient-to-hospital stages. It selects hospitals based on minimum expected door-to-needle time.	The work is primarily a simulation-based decision model rather than a complete multi-stakeholder platform and does not address request-level cascading fallback after ambulance rejection or timeout.
12	Schmid, 2012 [12]	This paper proposes an Approximate Dynamic Programming approach for dynamic ambulance relocation and dispatching under stochastic conditions. It jointly optimizes dispatch and relocation and reduces average response time from 4.60 to 4.01 minutes (12.89% improvement).	ADP faces the curse of dimensionality and requires state aggregation, creating a trade-off between computational complexity and solution accuracy. The study does not fully integrate real-time traffic data, advanced ML-based prediction, or continuous multi-stakeholder emergency coordination.

13	Tluli et al., 2025 [13]	This work proposes a dynamic redeployment system for Critical Care Response Vehicles using Deep Reinforcement Learning. It models redeployment as an MDP with a Deep Scoring Network and REINFORCE policy gradient.	The approach requires historical and operational data for training, uses REINFORCE with lower sample efficiency than PPO or SAC, and focuses on proactive redeployment rather than request-level cascading fallback.
14	Chu et al., 2023 [14]	This paper proposes a Deep Encoder Cross Network for ETA prediction using traffic, weather, road-speed, congestion, and other proxy features. It improves ETA prediction accuracy, particularly for late arrivals.	The model requires substantial training data and a relatively complex deep-learning architecture, while prediction quality depends on the quality and availability of proxy features.
15	Abid et al., 2024 [15]	This work develops an ambulance travel-time estimation framework using spatiotemporal data from SOS Alarm in Skåne, Sweden. It compares Polynomial Regression, XGBoost, and ANN, with ANN and XGBoost providing better estimation than Polynomial Regression.	The approach relies primarily on historical ambulance data rather than fully real-time traffic information, while Google Maps travel time introduces dependency on an external routing estimate. Weather and other real-time environmental conditions are not incorporated.

IV. PROPOSED SYSTEM

A. System Overview

Based on the research gap identified from the reviewed studies, UyirKappan is proposed as a unified emergency-response platform that integrates ambulance matching, real-time coordination, cascading fallback dispatch, and hospital resource visibility within a single workflow. The platform connects three client-facing components: a Bystander Mobile Application, a Driver/Ambulance Mobile Application, and a Hospital Dashboard through a centralized backend and shared database. Rather than treating ambulance dispatch as a single

request-and-assignment operation, the system maintains the emergency as a continuous workflow from request initiation to hospital arrival.

When an emergency request is submitted, the backend validates and records the request and initiates ambulance selection. Suitable ambulance candidates are identified using H3-based spatial filtering and subsequently evaluated using route-based cost information. The selected ambulance receives the assignment through the Driver Application. If the ambulance rejects the request or fails to respond within the configured response period, the system initiates cascading fallback and attempts reassignment using the remaining eligible candidates. Once an ambulance accepts the request, its location, status, and estimated time of arrival are continuously communicated to the relevant stakeholders. The Hospital Dashboard additionally provides visibility into incoming cases and hospital resource availability.

B. System Architecture

The proposed system follows a centralized architecture consisting of three application interfaces, a backend coordination layer, and a shared database. The Bystander Application is responsible for emergency request initiation and response tracking, while the Driver/Ambulance Application receives assignments and provides ambulance status and location updates. The Hospital Dashboard provides visibility into incoming emergencies and hospital resource availability.

The backend acts as the central coordination layer for request processing, ambulance matching, assignment management, fallback handling, and real-time information exchange. The Matching Engine first performs H3-based spatial candidate filtering and then applies Dijkstra-based route-cost evaluation to rank suitable ambulances. This approach avoids relying solely on geographic proximity, since the geographically nearest ambulance may not necessarily provide the most suitable route or response time.

When an assignment fails because of rejection or timeout, the fallback mechanism excludes the unsuccessful candidate and invokes the matching process again using the remaining eligible ambulances. After acceptance, ambulance location and response information are propagated through the backend to maintain a synchronized view across the Bystander Application and Hospital Dashboard. Thus, the architecture separates application interfaces from core coordination and decision-making while maintaining a centralized view of the emergency state.

The proposed architecture is designed to address the combined limitations identified in the reviewed studies rather than introducing a new routing algorithm in isolation. By integrating spatial candidate filtering, route-based ambulance selection, automatic fallback, and real-time coordination within a single workflow, UyirKappan provides a unified approach to emergency response. The architecture also maintains a shared emergency state across the participating stakeholders, enabling the response to continue from initial

request through ambulance arrival and hospital coordination without requiring separate manual coordination steps.

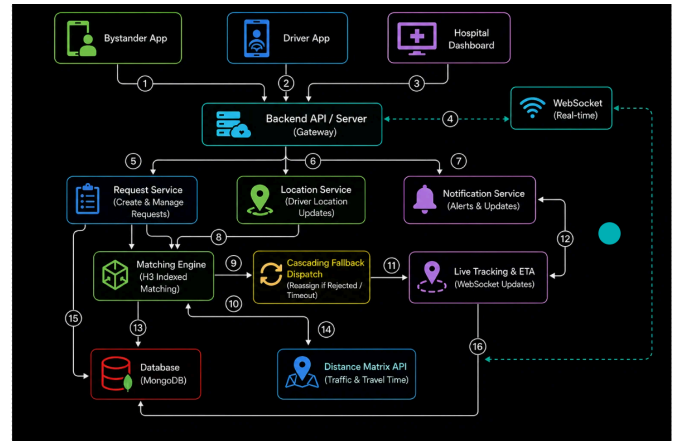


Fig. 1. Overall architecture of the proposed UyirKappan system.

V. IMPLEMENTATION

The proposed UyirKappan architecture has been realized as a working prototype consisting of a Bystander Mobile Application, Driver/Ambulance Mobile Application, and Hospital Dashboard connected through a centralized backend. The prototype demonstrates the major interactions identified from the research gap, including emergency request initiation, ambulance assignment, driver response, live tracking, hospital coordination, and fallback handling. The following screenshots provide visual evidence of the implemented interfaces and their roles within the emergency-response workflow.

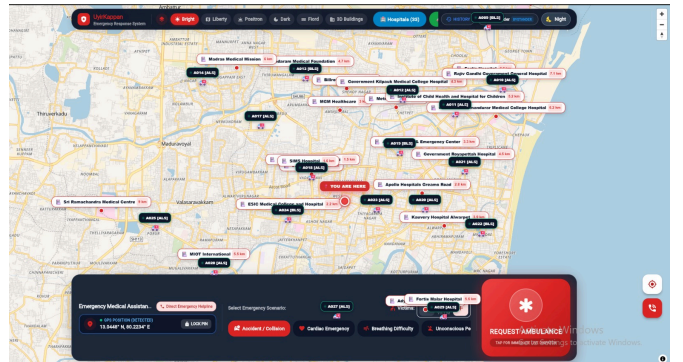


Fig. 2. Bystander App: Emergency request interface

The Bystander Application provides the interface for initiating an emergency request by capturing the user's location and emergency information. Following submission, the application provides response-status information and supports monitoring of the assigned ambulance.

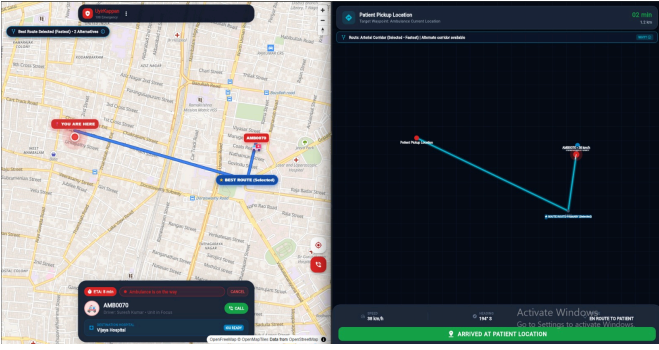


Fig. 3. Bystander App: Live tracking with ETA and route

The live-tracking interface provides the requester with the ambulance's current position, route, and estimated time of arrival, demonstrating the real-time coordination capability of the proposed platform.

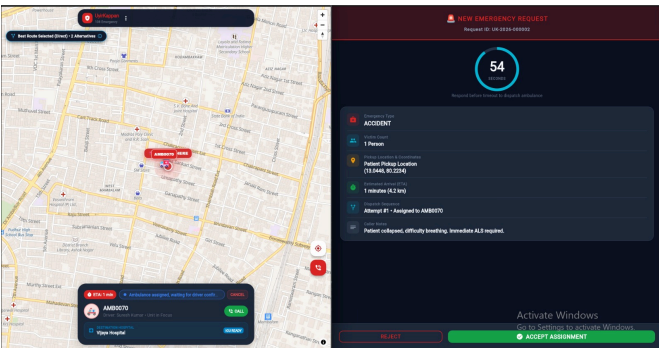


Fig. 4. Driver App: New emergency assignment with response timer

The Driver Application presents incoming emergency assignments and provides a defined response window for accepting or rejecting an assignment. Assignment failure can subsequently trigger the cascading fallback process described in the proposed system.

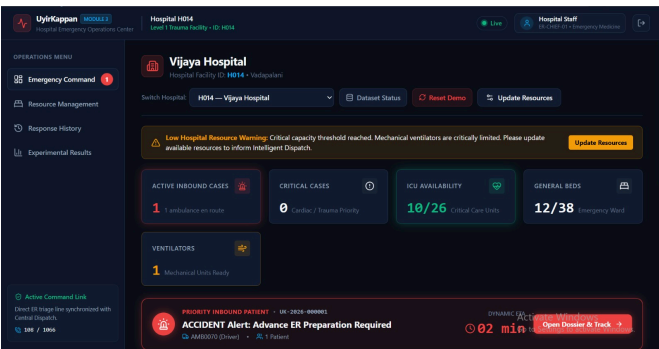


Fig. 5. Hospital Dashboard: Emergency command and inbound patient alert

The Hospital Dashboard provides hospital-side visibility of incoming emergency cases and relevant ambulance information, allowing hospital staff to prepare for the arriving patient and monitor available resources.

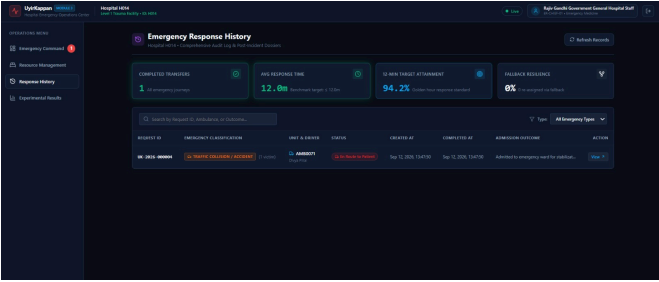


Fig. 6. Hospital Dashboard: Response history with metrics

The response-history interface provides visibility into completed emergency responses and associated response information, supporting monitoring and post-response analysis.

VI. CONCLUSION

This survey reviewed existing research on ambulance dispatch, hospital selection, driver-side coordination, backend infrastructure, and live tracking for emergency response, finding that these capabilities are typically developed in isolation rather than as part of a single coordinated workflow. Many existing systems rely on geographic proximity alone, address only one stage of the response chain, or leave recovery from a failed assignment unaddressed. In response to this gap, this paper presented UyirKappan, a unified emergency-response platform, along with a working prototype implementing intelligent ambulance matching, cascading fallback dispatch, real-time multi-stakeholder tracking, and hospital resource visibility across three connected client applications. The implementation demonstrates that these previously isolated capabilities can be integrated within a single deployable system, connecting the bystander, ambulance provider, and receiving hospital throughout the emergency lifecycle. Future work includes quantitative performance evaluation against nearest-ambulance dispatch, traffic-signal pre-emption for green corridor coordination, in-transit patient telemetry, and predictive ambulance availability using machine learning.

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